

Assignment 1

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Cougar Killsite Data Questions

Look at the years and months this data was collected to determine completeness of this data. You should also check the data dictionary. If someone asked you if cougars ate during the winter, how would you answer them?

Answers

The data was only collected 2009-2012 and 2015-2017 so there are gaps in years as well as the data was only collected between the months of March and October. This data set would not answer the question if cougars ate during the winter months as they focused “our field data collection on cougar predation behavior at times when cougars most frequently encounter black bears (i.e., bear active season). Thus, we only used kill-site investigation data for cougar kills made between 1 March and 31 October of each year”. So there was no data collection during winter months.

The data does have some consistency issues. PreyAgeClass has 14 different classifications and one example is where “Adult” and “adult” are two different groups. Being aware that there are overlapping ages as in “YOY (3-6 mo)” and “YOY (5-6 mo)”.

PreySpeciesSingle has consistency issues as it has 5 different identifiers for “Feral Horse”. This would need to be corrected before working more with the data set. Unless identifying wrong, there is also consistency issues with Scavenged - animal and the regular animal name.

```
library(readxl)
Killsites <- read_excel("Downloads/Cougar Killsites.xlsx")
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
unique(Killsites$PredationMonth)
```

```
## [1] "March"      "April"      "May"        "June"       "July"       "August"
## [7] "September" "October"
```

```
unique(Killsites$CougarID)
```

```
## [1] "57M"  "55F"  "53F"  "54F"  "52M"  "51M"  "60M"  "61M"
## [9] "59F"  "58M"  "64F"  "62F"  "68F"  "75F"  "71F"  "74M"
## [17] "77F"  "81F"  "84F"  "88F"  "85F"  "096F" "102M" "103F"
## [25] "101F" "36M"  "104F" "10345F" "10346F" "10645M" "10347F"
```

```
unique(Killsites$PreyAgeClass)
```

```
## [1] "Adult"
## [2] "Unknown"
## [3] "adult"
## [4] "YOY (0-3 mo)"
## [5] "YOY (3-6 mo)"
## [6] "YOY (unknown months, but likely 6-12 mo)"
## [7] "YOY (6-12 mo)"
## [8] "Yearling"
## [9] "YOY"
## [10] "YOY (0-1 mo)"
## [11] "YOY (2-3 mo)"
## [12] "YOY (unknown months, but guess 0-3 months)"
## [13] "YOY (5-6 mo)"
## [14] "YOY (unknown months, but likely 3-6 mo)"
```

```
unique(Killsites$PreySpeciesSingle)
```

```
## [1] "Feral Horse"      "badger"
## [3] "feral Horse"      "mule deer"
## [5] "small mammal"     "domestic dog"
## [7] "Scavenged - domestic cow" "coyote"
## [9] "porcupine"        "Feral Horsse"
## [11] "Domestic Cattle"  "feral horse"
## [13] "Unknown Yearling Ungulate" "Domestic Sheep"
## [15] "pronghorn"        "Scavenged - mule deer"
## [17] "Beaver"           "bighorn sheep"
## [19] "Scavenged - feral horse" "Black Bear"
## [21] "Unknown YOY Ungulate" "Red Fox"
## [23] "Bird"             "bobcat"
```